

level 1B: Find the Fake coin.

Enigma no. = 3706

(github username = Harsith902)

given info:

- ① The Fake coin is slightly heavier than normal one.
- ② we have a weighing Balance with 2 pans.

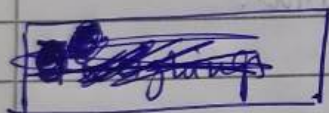
my coins

$$n = 10 + \text{Remainder of } \left(\frac{3706}{90} \right)$$

$$n = 10 + 16$$

$$\boxed{n = 26}$$

Que 1. Guess how many weighings for 27 coins?

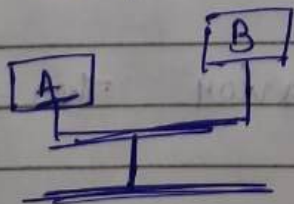


$\Rightarrow$ 4 weighings

Que 2. How many weighings for 27 coins, in worst case.

① let's divide 27 coins as 13, 13, 1

let 1 coin rest and put weigh the 13, 13 coins in the pan.



//_

(i) case ①

if weight of A = wt (B)

then the Fake coin is the one which was kept aside.

(ii) case ②

If one of the pan is heavier then the fake coin is among them.

let (A) pan be heavier so we keep all coins of (B) aside now.

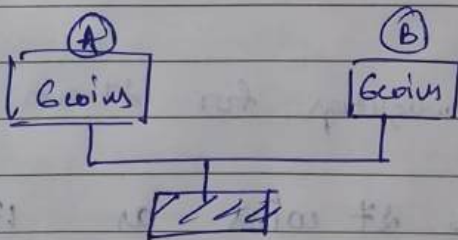
A $\Rightarrow$ 13 coins.

Repeat the same.

divide as 6, 6, 1 coin.

keep aside 1 coin.

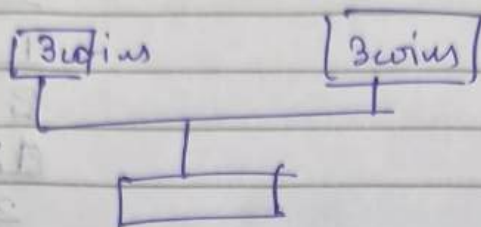
(a) If



If Both balanced then the coin kept aside is Fake.

⑥ If one pan is heavier then the fake is among them.

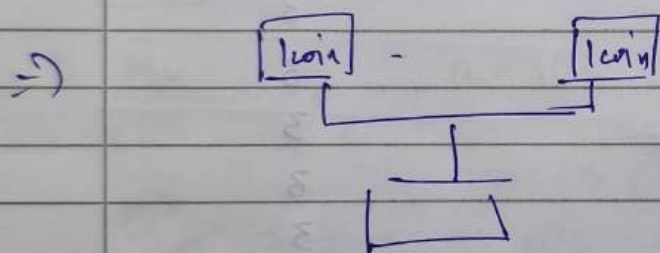
we divide $6 \rightarrow 3, 3$



~~if it is~~

① since the fake one lies among them
one pan would be heavier so we repeat.

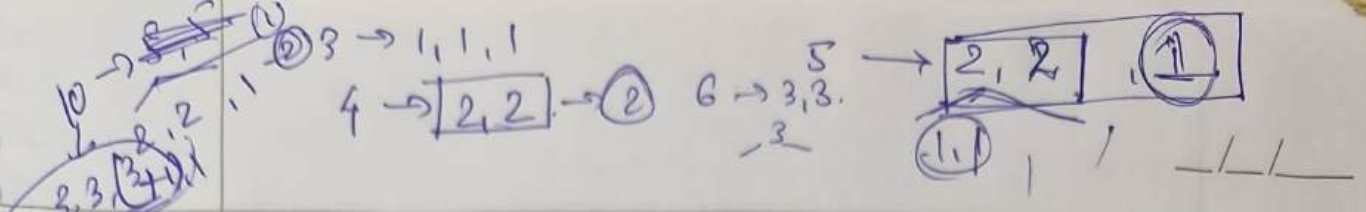
$3 \rightarrow 1, 1, 1$



① If Balanced then the coin kept aside is fake

② If unbalance then the fake coin is the heavier one.

Therefore, in the worst case we would require 4 weighings.



8.

Coins

weighing (min)

2

1

3

1

4

2

5

2

6

2

7

2

8

2

9

2

10

3

11

3

12

3

13

3

14

3

15

3

16

3

17

3

18

3

19

3

20

3

21

3

22

3

23

3

24

3

25

3

26

3

27

3

4. Where does the ~~not~~ no. of weighing go up by one.

the no. of weighing are changing at.

3 coins, 9 coins, 27 coins.

so after every ~~weighing~~

~~so weighing~~

~~so~~

so ~~for~~ x weighings can handle
till 3^x coins

then the weighings increases by 1.

so in that case.

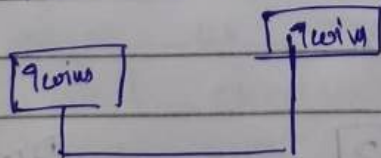
for 28 coins the fewest weighing would be 4.

5. My coin: $n = 10$ + Remain $\left(\frac{3706}{95} \right)$

$$n = 10 + 16 = 26$$

now I have 26 coins so I would divide it
as ~~8, 8, 8~~ 9, 9, 8 coins

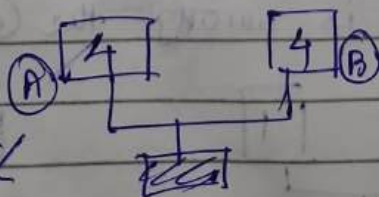
① Take 9 coins and 9 coins in weighing Pan



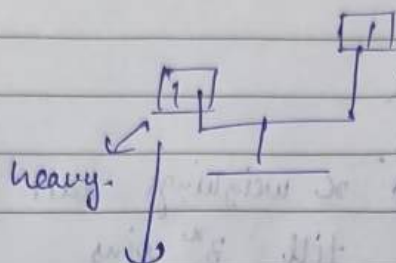
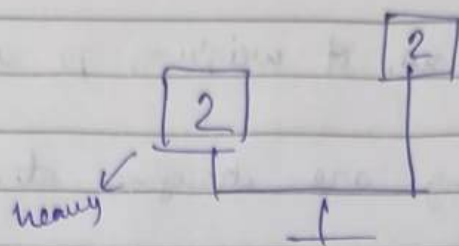
→ ① If Balanced

The fake coin is ~~in~~ among the 8 coins

80



heavy



this is fake one.

so by dividing the coins of heavier pan and weighing would lead to finding fake coin so this case

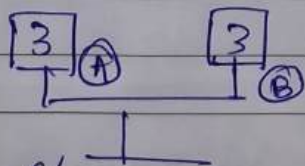
The no. of weighing is 4
~~the weighing is~~

(ii) not balanced

The fake coin is in among heavier parts of 9 coins we divide.

3, 3, 3.

②



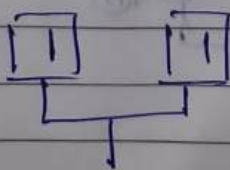
3 coins

③

~~if~~ Balanced

⊗ Fake is among the ②

so



1 coin

//_

The heavier coin is fake if balanced
the one kept aside is fake

no. of weighing = 3

~~not a~~

~~similarly~~ ~~the~~ if the we

similarly if the 3 3 was not balanced

same would done.

Therefore. The fewest weightings possible is 3

But this is not same as what we did by
~~dividing~~ ^{parting} in 2 parts. (4 weightings)

Que 6.

① Splitting in 3 parts beat 2 parts because
it gives us more wider based approach
and will eliminate more useless coins
that we won't need.

②:

② Every weighing has 3 results

- Right pan heavy
- left pan heavy
- Balanced.